

```
from google.colab import files
uploaded = files.upload()
```

t1.mp4

t1.mp4(video/mp4) - 4776626 bytes, last modified: 4/7/2026 - 100% done
Saving t1.mp4 to t1.mp4

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Get uploaded video filename
filename = list(uploaded.keys())[0]

# Open video file
cap = cv2.VideoCapture(filename)

if not cap.isOpened():
    print("Error: Could not open video file")
else:
    # Read first frame
    ret, first_frame = cap.read()
    if not ret:
        print("Error: Could not read video")
    else:
        prev_gray = cv2.cvtColor(first_frame, cv2.COLOR_BGR2GRAY)
        prev_gray = cv2.GaussianBlur(prev_gray, (15, 15), 0)

        print("Motion Tracking Started...")

        frame_count = 0

        while True:
            ret, frame = cap.read()
            if not ret:
                break

            gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
            gray = cv2.GaussianBlur(gray, (15, 15), 0)

            # Frame difference
            frame_diff = cv2.absdiff(prev_gray, gray)

            # Threshold
            thresh = cv2.threshold(frame_diff, 15, 255, cv2.THRESH_BINARY)[1]
            thresh = cv2.dilate(thresh, None, iterations=3)

            # Find contours
```

```
contours, _ = cv2.findContours(thresh.copy(), cv2.RETR_EXTERNAL, cv2.CV_CHAIN_APPROX_SIMPLE)

motion_detected = False

for contour in contours:
    if cv2.contourArea(contour) > 200:
        motion_detected = True
        x, y, w, h = cv2.boundingRect(contour)
        cv2.rectangle(frame, (x, y), (x+w, y+h), (0, 255, 0), 2)

# Status text
status = "MOTION DETECTED" if motion_detected else "No Motion"
color = (0, 255, 0) if motion_detected else (0, 0, 255)

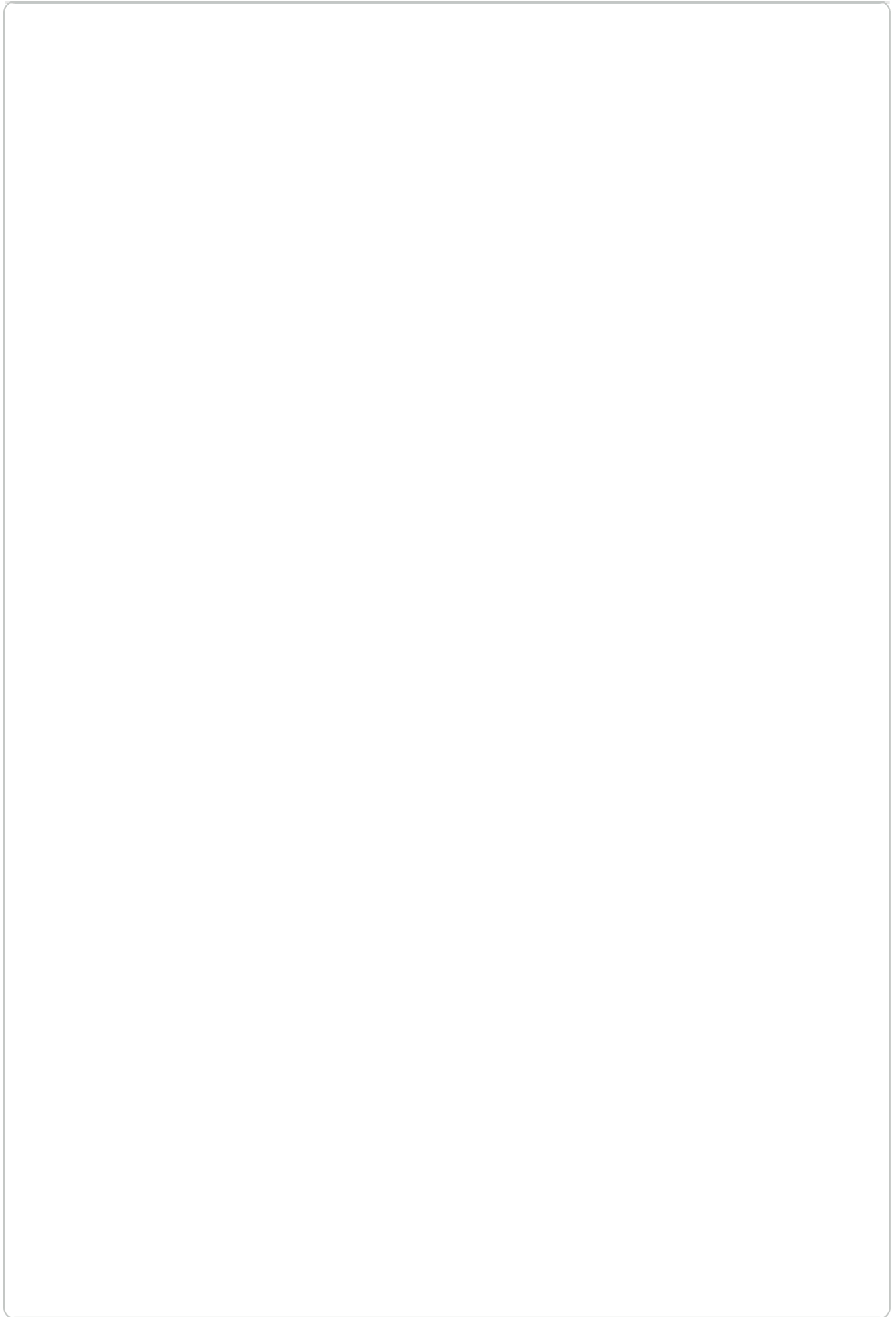
cv2.putText(frame, status, (10, 30),
            cv2.FONT_HERSHEY_SIMPLEX, 1, color, 2)

# Convert to RGB for display
frame_rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)

# Show every 10th frame (to reduce lag in Colab)
if frame_count % 10 == 0:
    plt.imshow(frame_rgb)
    plt.title(status)
    plt.axis("off")
    plt.show()

prev_gray = gray
frame_count += 1

cap.release()
print("Motion Tracking Finished")
```



Motion Tracking Started...

MOTION DETECTED



MOTION DETECTED



MOTION DETECTED

